# Numerical Simulation of Atmospheric Pollutant Dispersion in an Urban Street Canyon Using CFD Methods

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## Abstract

This report presents numerical investigation of atmospheric pollutant dispersion within an idealized urban street canyon, validated against the benchmark study by Salim (2011). The study focuses on the transport of sulfur hexafluoride ($SF_6$) under perpendicular approaching wind conditions. Calculations are performed using the steady state Reynolds Averaged Navier Stokes (RANS) framework coupled with the advection diffusion method for species transport.

Two turbulence closure models are evaluated and compared: the Standard $k_\epsilon$ and the $k_\omega$ Shear Stress Transport (SST) model. To accurately replicate the atmospheric boundary layer, User Defined Functions (UDFs) were implemented to describe vertical inlet profiles for mean velocity, as well as profiles for turbulent kinetic energy ($k$) and specific dissipation rate ($\omega$).

The comparative analysis identifies distinct differences between the two RANS models in predicting structure of vortices and the resulting pollutant accumulation zones near the leeward, windward walls (Wall A, Wall B).

# Contents

# Chapter 1

# Introduction

## 1.1 Background and Motivation

Air pollution in urban environments remains a critical challenge for health, safety, and sustainable urban planning. Within idealized street canyons, the complex interaction between the atmospheric boundary layer and building geometries significantly alters local flow patterns, often reduces natural ventilation. In these canyons, the wind often forms large, stable rotating vortices. These vortices act like a trap, preventing fresh air from entering and causing pollutants to accumulate at the ground level where people breathe. Because of this limited ventilation, gas concentrations can reach dangerously high levels near the street surface.

Understanding these mechanisms is essential, as poorly ventilated places pose a direct risk to inhabitants. While experimental wind tunnel studies provide high quality data, Computational Fluid Dynamics (CFD) offers a cost effective and detailed alternative for analyzing transport of pollutant gases, such as sulfur hexafluoride ($SF_6$).

In this study, the motivation lies on evaluating predictive capabilities of the steady state Reynolds Averaged Navier Stokes (RANS) framework. Specifically, the performance of the Standard $k_\epsilon$ model is compared against $k_\omega$ Shear Stress Transport (SST) model. By implementing specific User Defined Functions (UDFs) to replicate the benchmark conditions of Salim (2011).

The primary motivation for this study is to perform validation of the numerical setup against the wind tunnel experiments of Salim et al. (2011). By matching these experimental results, we ensure that the chosen mesh strategy, turbulence models ($k_\epsilon$ and $k_\omega$ SST), and boundary conditions accurately represent physical reality. This validation serves as a necessary guarantee, providing a reliable and proven CFD framework that will be applied to more complex urban geometries in future research.

## 1.2 Problem Statement

This study considers an idealized urban street canyon with an aspect ratio of $W/H = 1$, where both the street width ($W$) and building height ($H$) are set to 18m. This configuration is based on the 1:150 scale wind tunnel benchmark by Salim et al. (2011).

The pollutant, sulfur hexafluoride ($SF_6$), is released from a ground level line source with a length of 216 m ($12H$) and is transported by a steady atmospheric boundary layer flow. The primary objective is to evaluate the accuracy of the RANS numerical setup by comparing the predicted flow patterns and pollutant concentration fields with the experimental data.

## 1.3 Sulfur Hexafluoride ($SF_6$) as a Pollutant Gas

In this work, sulfur hexafluoride ($SF_6$) is used as a tracer gas for pollutant dispersion modeling. It is released from street-level line sources that represent traffic-related emissions. This gas is widely used in dispersion studies because it allows researchers to track the transport of pollutants under controlled conditions. By analyzing the concentration field of $SF_6$, it is possible to evaluate ventilation performance and identify regions where pollutants tend to accumulate.

## 1.4 Objectives

The primary goal of this research is to establish a validated CFD framework for urban dispersion studies. The specific objectives are:

- To construct a 3D geometry of an idealized urban street canyon with a full-scale aspect ratio ($W/H = 1$);
- To implement a robust CFD setup in ANSYS Fluent for steady state RANS simulations, integrating species transport for pollutant gas;
- To define realistic atmospheric boundary layer conditions using User Defined Functions (UDFs) for velocity, $k$, $\epsilon$, and $\omega$ profiles;
- To compare Standard $k_\epsilon$ and $k_\omega$ SST turbulence models;
- To validate the numerical results against the experimental data of Salim (2011) to ensure the physical accuracy of the model;
- To identify pollutant accumulation zones.

# Chapter 2

# Geometry and Mesh

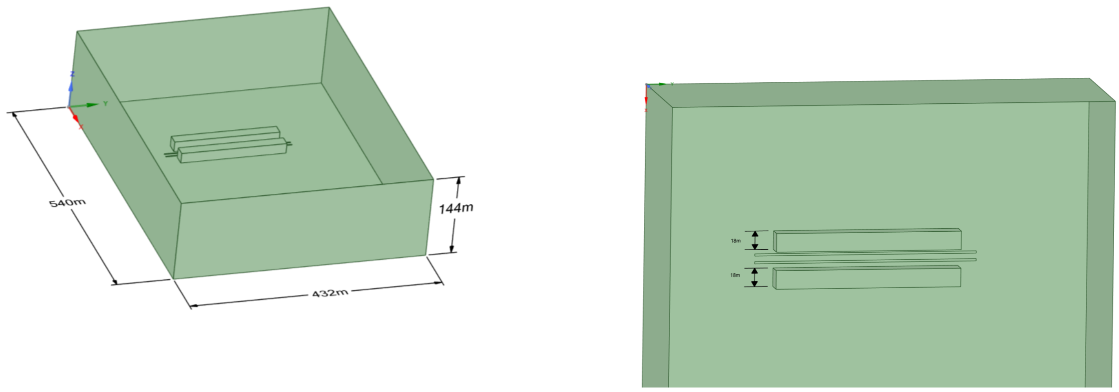
## 2.1 Geometry and Computational Domain

The numerical investigation is based on an idealized Urban Street Canyon (USC) configuration. The geometry is defined by a unity aspect ratio, where the street width ($W$) and building height ($H$) are both set to 18 m ($W/H = 1$). While the original experimental wind tunnel (WT) data from the University of Karlsruhe used a 1:150 scale ($H = 0.12$ m).

The flow domain and its respective boundary conditions are defined as follows:

- **Main Air Inlet:** Located at $8H$ upstream of the first building.
- **Outlet:** Located at $19H$ downstream to allow for full wake development.
- **Upper Boundary (Top):** A symmetry boundary condition (zero shear slip) was applied at a height of $8H$ from the floor.
- **Sides:** Symmetry conditions were applied to the lateral boundaries, located at a distance of $7H$ from the building edges.
- **Pollutant Sources:** Two line sources, each 216m ($12H$) in length, were placed at the canyon floor to represent traffic emissions, consistent with $SF_6$.

The detailed dimensions of the computational domain and the nested subdomains, designed for mesh refinement around the canyon, are illustrated in Figure 2.1.




(a) Top view of the domain

(b) Side view of the canyon and subdomains

Figure 2.1: Computational domain setup: (a) layout and subdomains; (b) applied boundary conditions and line source locations.

## 2.2 Computational Mesh

To achieve a balance between computational cost and spatial resolution, a hybrid hexa-dominant mesh was generated. The domain was discretized using a multi-zone approach, in which the region surrounding the street canyon and the pollutant sources was heavily refined. The mesh parameters are summarized as follows:

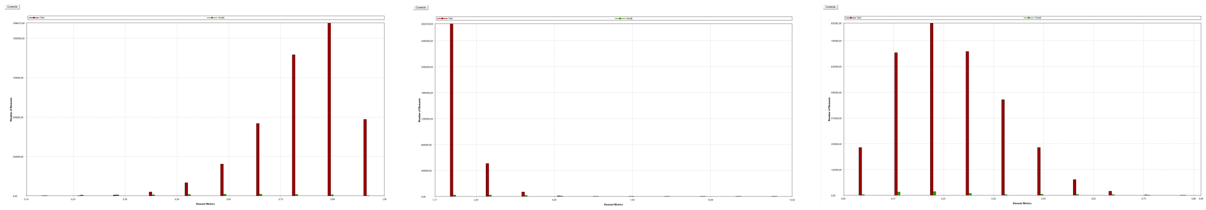
- **Global Mesh Size:** $H/4.5 \approx 4$ m, ensuring a baseline resolution for the far-field flow.
- **Refinement Zone (Sub-mesh):** Within the street canyon and near building surfaces, the element size was reduced to $H/9 \approx 2$ m.
- **Boundary Layer Treatment:** To accurately resolve the viscous sublayer and pressure-driven separation at the building corners, 5 inflation layers were applied to all solid wall boundaries.
- **Mesh Statistics:** The final discretized domain consists of approximately 591,495 nodes and 3,388,605 elements.

## 2.3 Mesh Quality Assessment

The accuracy and stability of the numerical solution are directly influenced by the quality of the computational mesh. In this study, three primary metrics were utilized to evaluate the mesh: Element Quality, Aspect Ratio, and Skewness.

- **Element Quality:** This composite metric evaluates the shape of the element on a scale from 0 to 1, where 1 represents a perfect equilateral cell and 0 indicates a degenerate cell with zero or negative volume. For 3D elements, it is calculated based on the ratio of the volume to the square root of the cube of the sum of the square of the edge lengths.
- **Aspect Ratio:** Defined as the ratio of the longest edge length to the shortest edge length of an element. High aspect ratios can lead to numerical instability, particularly in regions with high flow gradients; however, values observed within the boundary layer (inflation layers) remain acceptable for the selected numerical setup.
- **Skewness:** This metric measures how close an element is to its ideal equilateral form. A skewness of 0 represents an ideal cell, while values approaching 1 indicate highly distorted elements that may hinder convergence.

The statistical distribution of these metrics is illustrated in Figure 2.2. The mean values recorded—0.77 for Element Quality, 1.83 for Aspect Ratio, and 0.22 for Skewness—demonstrate a high-quality discretization suitable for steady species transport analysis.

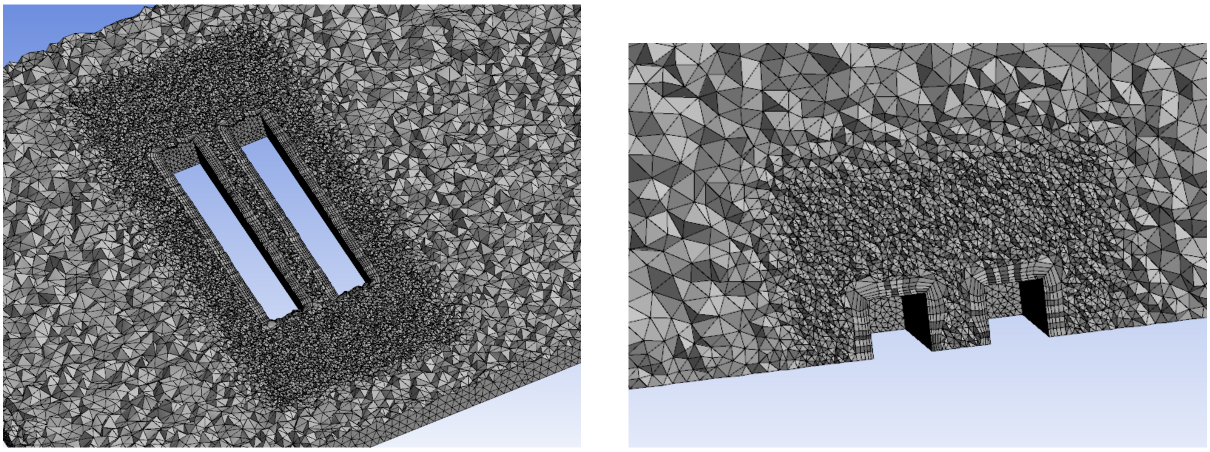


(a) Element Quality (b) Aspect Ratio (c) Skewness

Figure 2.2: Mesh quality histograms: (a) Element Quality distribution (Mean: 0.77); (b) Aspect Ratio distribution (Mean: 1.83); (c) Skewness distribution (Mean: 0.22).

The statistical distributions of Element Quality, Aspect Ratio, and Skewness are illustrated in Figure 2.2. The corresponding mean values are 0.77, 1.83, and 0.22, respectively, indicating good overall mesh quality.

Although the maximum aspect ratio reaches 31.65 near the walls due to the boundary-layer inflation layers, these extreme values are highly localized. In the critical flow region, most elements retain low skewness and acceptable shape characteristics, which helps reduce numerical diffusion during the steady species transport simulation.



(a) Top view (b) Side view

Figure 2.3: Cross-sectional views of the computational mesh: (a) Horizontal plane (top view) (b) Vertical plane (side view)

# Chapter 3

# Numerical Setup and Boundary Conditions

## 3.1 Governing Equations

The fluid flow is governed by the steady-state 3D Reynolds-Averaged Navier-Stokes (RANS) equations. Since the simulation is performed under steady conditions, the time-dependent terms are neglected. The mass conservation and momentum equations are expressed as follows:

$$\nabla \cdot (\rho \mathbf{u}) = 0 \tag{3.1}$$

$$\nabla \cdot (\rho \mathbf{u}\mathbf{u}) = -\nabla p + \nabla \cdot (\tau) + \rho \mathbf{g} \tag{3.2}$$

For the pollutant dispersion, a species transport model without chemical reactions was employed. The local mass fraction of the pollutant ($SF_6$), denoted as $Y_i$, is predicted through the solution of a convection-diffusion equation:

$$\nabla \cdot (\rho \mathbf{u} Y_i) = -\nabla \cdot \mathbf{J}_i + S_i \tag{3.3}$$

where $\mathbf{J}_i$ is the diffusion flux of the species.

## 3.2 Material Properties and Species Transport

The dispersion study involves a mixture of air and Sulfur hexafluoride ($SF_6$) as a tracer gas. To accurately simulate the buoyancy effects, the following properties were implemented:

- **Tracer Gas ($SF_6$):** The molecular weight is set to 146.06 kg/kmol, making it approximately five times heavier than air. The density is modeled using the ideal-gas law, and the viscosity is defined as $1.53 \times 10^{-5}$ Pa·s.
- **Mixture Properties:** The background air viscosity is $1.7894 \times 10^{-5}$ kg/m·s. The mass diffusivity for the $SF_6$-air mixture is specified as $9.5 \times 10^{-6}$ m$^2$/s.
- **Turbulent Schmidt Number:** Following the standard recommendations for atmospheric dispersion, the Turbulent Schmidt Number ($Sc_t$) is set to 0.7.

## 3.3 Boundary Conditions and UDF

To simulate a realistic Atmospheric Boundary Layer (ABL) under neutral stability conditions, the inlet velocity, turbulent kinetic energy ($k$), and dissipation rate ($\varepsilon$) were prescribed using a User-Defined Function (UDF). This approach ensures an equilibrium ABL, preventing the artificial decay of turbulence as the flow traverses the computational domain. The vertical profiles are defined by the following analytical expressions:

$$u(z) = 4.7\left(\frac{z}{H}\right)^{0.3} \tag{3.4}$$

$$k = \frac{u_*^2}{\sqrt{C_\mu}}\left(1 - \frac{z}{\delta}\right) \tag{3.5}$$

$$\varepsilon = \frac{u_*^3}{\kappa z}\left(1 - \frac{z}{\delta}\right) \tag{3.6}$$

$$\omega = \frac{\varepsilon}{C_\mu k} = \frac{u_*}{\kappa \sqrt{C_\mu} z} \tag{3.7}$$

Where the physical parameters and constants are defined as follows:

- $H = 18$ m is the reference building height.
- $\delta = 144$ m is the boundary layer depth, scaled to the full-size geometry.
- $u_* = 0.54$ m/s is the friction velocity.
- $\kappa = 0.4$ is the von Karman constant.
- $C_\mu = 0.09$ is the standard empirical constant for the $k - \varepsilon$ model.

Listing 3.1: UDF

```
#include "udf.h"

#define U_STAR 0.54
#define DELTA 144.0
#define KAPPA 0.4
#define CMU 0.09

DEFINE_PROFILE(velocity_inlet, thread, position)
{
  real x[ND_ND], z;
  face_t f;
  begin_f_loop(f, thread)
  {
    F_CENTROID(x, f, thread);
    z = x[2];
    if (z < 0.01) z = 0.01;
    F_PROFILE(f, thread, position) = 4.7 * pow(z / 18.0, 0.3);
  }
  end_f_loop(f, thread)
}

DEFINE_PROFILE(k_profile, thread, position)
{
  real x[ND_ND], z;
  face_t f;
  begin_f_loop(f, thread)
  {
    F_CENTROID(x, f, thread);
    z = x[2];
    if (z > DELTA) z = DELTA;
    F_PROFILE(f, thread, position) = (pow(U_STAR, 2) / sqrt(CMU)) *
        (1.0 - z/DELTA);
  }
  end_f_loop(f, thread)
}

DEFINE_PROFILE(eps_profile, thread, position)
```

```
{
  real x[ND_ND], z;
  face_t f;
  begin_f_loop(f, thread)
  {
    F_CENTROID(x, f, thread);
    z = x[2];
    if (z < 0.1) z = 0.1;
    if (z > DELTA) z = DELTA * 0.99;

    F_PROFILE(f, thread, position) = (pow(U_STAR, 3) / (KAPPA * z)) *
       (1.0 - z/DELTA);
  }
  end_f_loop(f, thread)
}

DEFINE_PROFILE(omega_profile, thread, index)
{
    real x[ND_ND];
    real z;
    face_t f;

    begin_f_loop(f, thread)
    {
        F_CENTROID(x, f, thread);
        z = x[2];

        F_PROFILE(f, thread, index) = U_STAR / (KARMAN * sqrt(CMU) * (z
           + 0.0033));
    }
    end_f_loop(f, thread)
}
```

The boundary conditions were defined to accurately replicate the atmospheric boundary layer and the specific pollutant emission source:

- **Main Inlet:** A velocity-inlet profile implemented via UDF with $u_* = 0.54$ m/s and a boundary layer depth $\delta = 144$ m (scaled to full-size geometry).
- **Pollution Source:** A *mass-flow-inlet* was used for the $SF_6$ release with a rate of 0.01 kg/s. The turbulence parameters at the source were set to a 2% intensity and a turbulent viscosity ratio of 2.5.
- **Outlet:** A pressure-outlet boundary condition with radial equilibrium pressure distribution was applied to allow the flow to leave the computational domain.
- **Walls:** No-slip conditions were applied to all building and ground surfaces.

## 3.4 Solver Configuration

The numerical solution was obtained using the following settings in ANSYS Fluent:

- **Solver Type:** Pressure-based steady-state solver.
- **Pressure-Velocity Coupling:** SIMPLE.
- **Gravity:** Enabled in the negative Z direction ($-9.81$ m/s$^2$).

- **Turbulence Model:** Standard $k - \varepsilon$ model with $C_\mu = 0.09$, $C_{1\varepsilon} = 1.44$, and $C_{2\varepsilon} = 1.92$.
- **Discretization Schemes:** Second-order upwind for momentum and species transport.

The numerical solution for the $k_omega$ SST model was obtained using the following settings in ANSYS Fluent to ensure superior capturing of flow separation and near-wall gradients:

- **Solver Type:** Pressure-based, steady-state, implicit solver.
- **Pressure-Velocity Coupling:** *Coupled* (or *SIMPLEC*) scheme was employed to enhance pressure-velocity stability in the complex wake regions of the urban canopy.
- **Turbulence Model:** $k - \omega$ Shear Stress Transport (SST).
- **Turbulence Constants:** Standard SST closure coefficients were applied ($\beta_i^* = 0.09$, $\alpha_1 = 5/9$, $\beta_1 = 0.075$).
- **Discretization Schemes:**
  - **Pressure:** Second-order.
  - **Momentum:** Second-order upwind
  - **Turbulent Kinetic Energy ($k$):** Second-order upwind.
  - **Specific Dissipation Rate ($\omega$):** Second-order upwind.
  - **Species Transport:** Second-order upwind for high-fidelity pollutant distribution.
- **Convergence Monitoring:** A convergence threshold of $10^{-5}$ for all residuals was enforced. Additionally, the mass fraction of $SF_6$ at key monitoring points was tracked to ensure the solution reached a stable plateau.

## 3.5 Solver Stability and Convergence

To ensure stable convergence and prevent numerical oscillations during the steady-state iterations, the Under-Relaxation Factors (URF) were optimized. Higher values were assigned to the species and energy equations to prioritize their convergence after the flow field became partially developed.

Table 3.1: Applied Under-Relaxation Factors

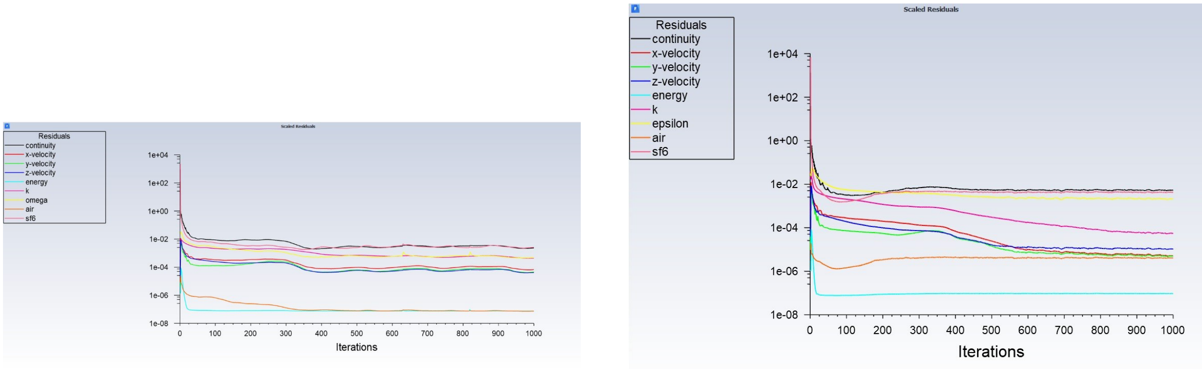
| **Variable** | **Factor Value** |
|---|---|
| Pressure | 0.3 |
| Momentum | 0.7 |
| Turbulent Kinetic Energy ($k$) | 0.8 |
| Dissipation Rate ($\varepsilon$) | 0.8 |
| Air (Mass Fraction) | 0.9 |
| $SF_6$ (Mass Fraction) | 0.9 |
| Energy | 1.0 |

The solution was considered converged when the scaled residuals for all equations reached a threshold of $1 \times 10^{-5}$. In addition to residuals, the mass fraction of $SF_6$ at the outlet was monitored to ensure it reached a steady-state value.

# Chapter 4

# Results

## 4.1 Convergence graphs

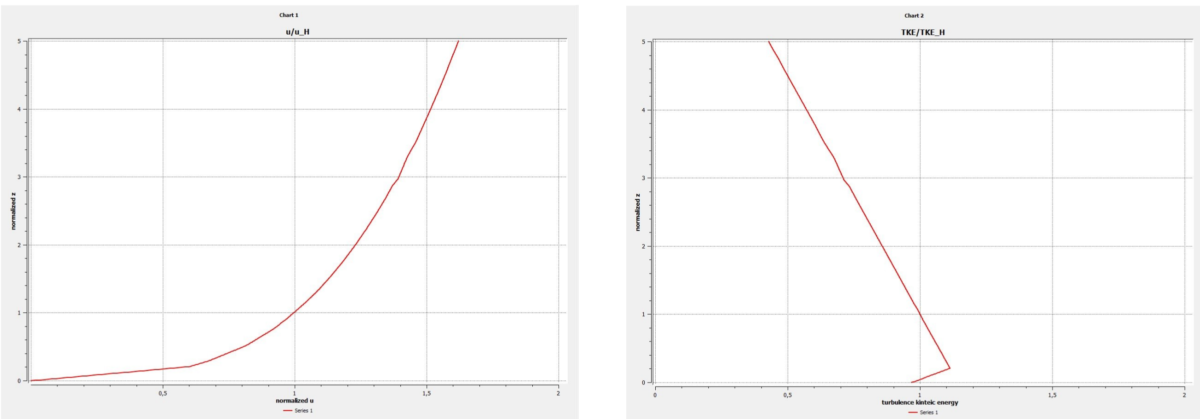



(a) Standard $k-\varepsilon$ model　　(b) $k-\omega$ SST model

Figure 4.1: Residual convergence history for the two turbulence models: (a) stable decline of residuals for $k-\varepsilon$; (b) convergence behavior for the $k-\omega$ SST model.

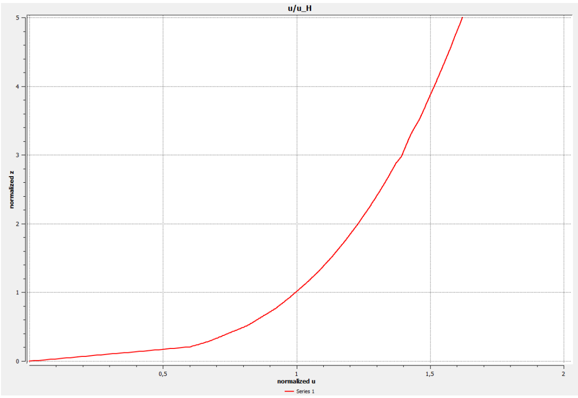
## 4.2 Inlet Boundary Layer Validation

The vertical profiles of the normalized velocity and turbulent kinetic energy (TKE) were extracted at the inlet boundary to verify the implementation of the User-Defined Function (UDF) and ensure the stability of the atmospheric boundary layer (ABL).

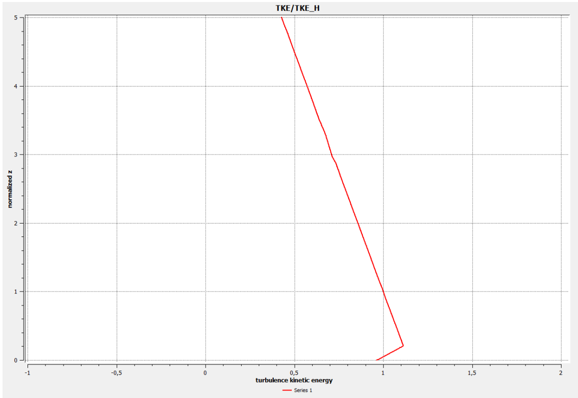



(a) Normalized velocity $u/u_H$　　(b) Normalized TKE $k/k_H$

Figure 4.2

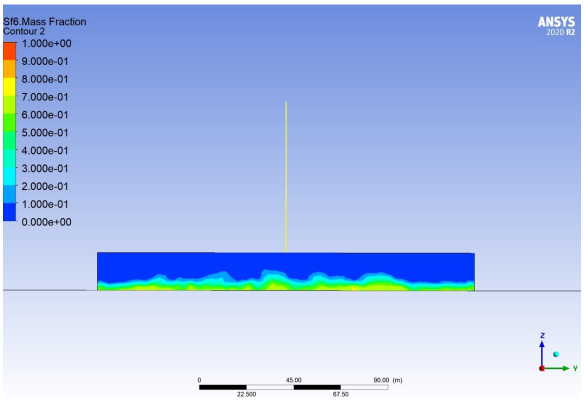



(a) Normalized velocity $u/u_H$




(b) Normalized TKE $k/k_H$

Figure 4.3




(a)

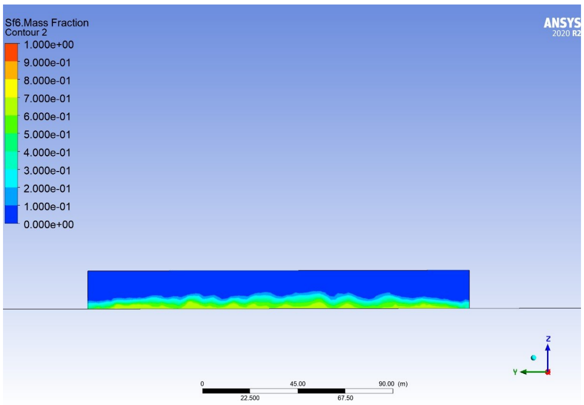



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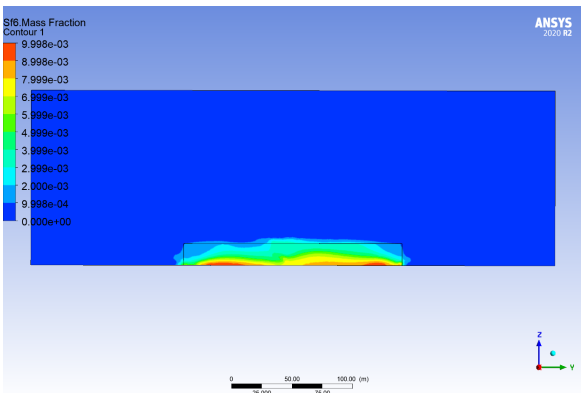
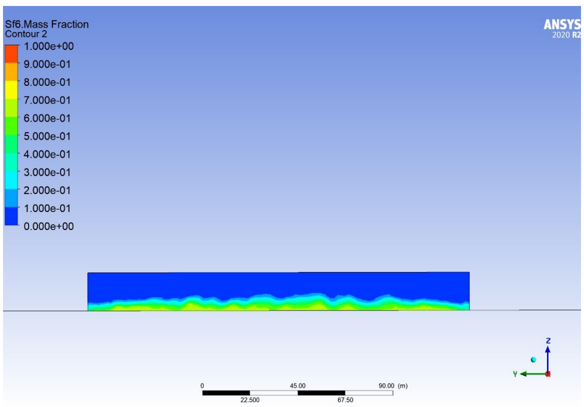
Figure 4.4




(a)




(b)

Figure 4.5